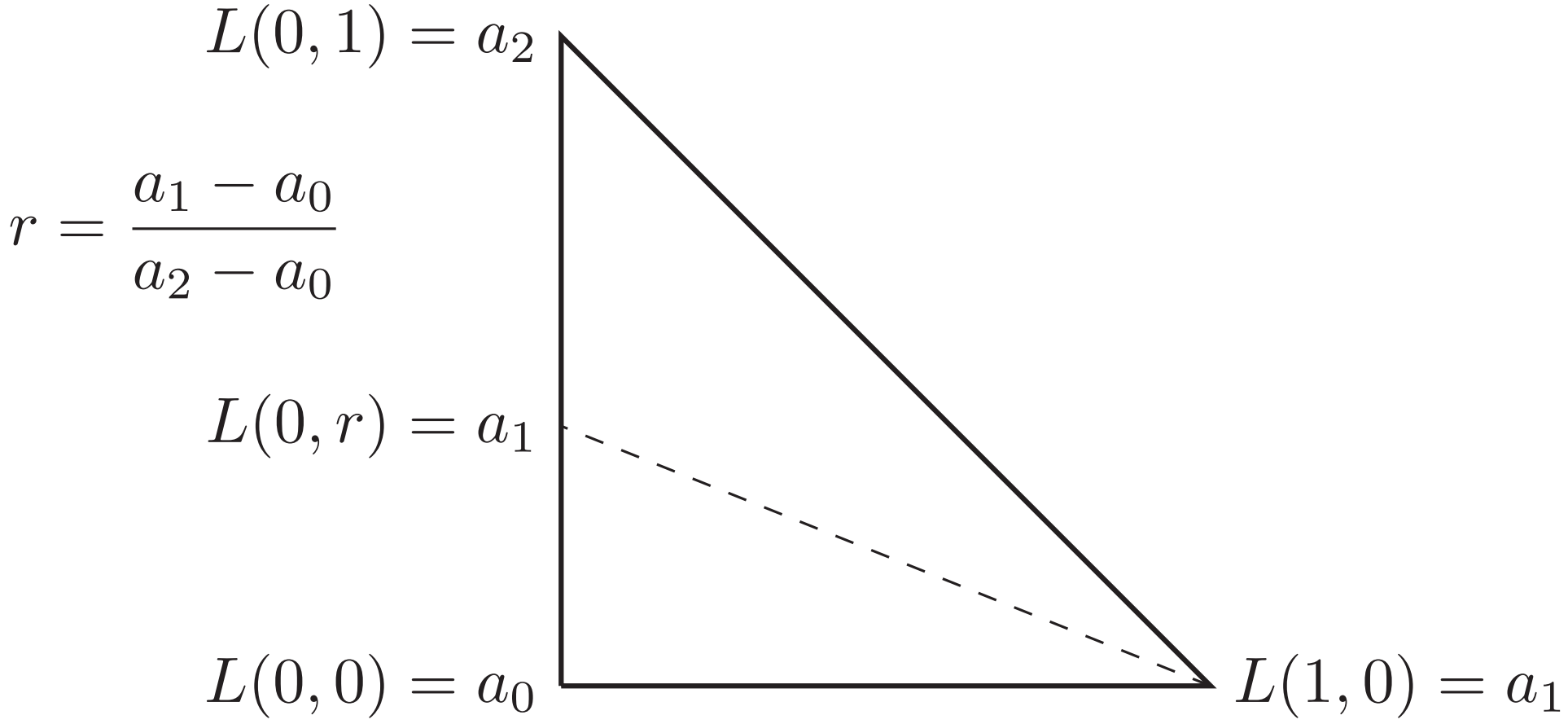

$L(0,1) = a_2$
$r = \dfrac{a_1 - a_0}{a_2 - a_0}$
$L(0,r) = a_1$
$L(0,0) = a_0$
$L(1,0) = a_1$